

STANDARD OPERATING PROCEDURE

In/Out Attendance and Wage Automation System

Document Control

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1 PURPOSE

This Standard Operating Procedure (SOP) defines the end-to-end process for:

- Recording employee attendance (IN/OUT punches) using the In/Out Attendance and Wage Automation System.
- Capturing location and device information to enforce geofence and device-based security.
- Calculating daily wages, including normal and overtime hours, and generating monthly summaries for payroll processing.

The objective of this SOP is to ensure that every user, regardless of department or experience level, follows a consistent, error-free process that produces accurate attendance and wage data in the target Google Sheet.

2 SCOPE

This SOP applies to:

- All employees required to mark attendance via the In/Out Attendance web application.
- HR and Operations staff responsible for reviewing attendance, validating anomalies, and approving wage calculations.
- Finance/Payroll team members who use monthly summaries for salary processing.

This SOP does **not** cover:

- Direct modification of backend code (Apps Script) or frontend HTML/JavaScript.
- Changes to geofence configuration or device security logic (handled by the System Administrator / Senior Software Developer).
- Broader HR policies for leave, absence, and disciplinary action beyond attendance capture and wage calculation.

3 DEFINITIONS AND ABBREVIATIONS

Table 1: Definitions and Abbreviations

Term / Abbreviation	Definition
Web App	Google Apps Script-based attendance interface used for IN/OUT punches.
Target Sheet	Central Google Spreadsheet where attendance and wage data is written.
Employees Sheet	Configuration sheet containing employee names, hourly rates, and status.
Monthly Sheet	Per-month sheet (MMM-yyyy) storing daily punches and monthly summary rows.
IN	Attendance status representing start of work or shift.

Term / Abbreviation	Definition
OUT	Attendance status representing end of work or shift.
Geofence	Configured circular area (center latitude/longitude and radius) around office premises used to validate location.
Device ID	Persistent identifier stored in browser localStorage and sent to backend for device anomaly tracking.
Device Anomaly	Condition where current device differs from historically established device beyond a defined threshold.
Normal Hours	Daily working hours up to configured limit (e.g., 8 hours).
Overtime Hours	Daily working hours exceeding the normal hours limit.
Total Wage	Calculated wage for a given OUT entry, based on normal and overtime hours and hourly rate.
FY	Financial Year (April to March).

4 ROLES AND RESPONSIBILITIES

Table 2: Roles and Responsibilities

Role	Responsibility
Employee	Use the Web App to mark IN/OUT attendance; ensure geolocation permissions are granted and device is stable.
HR Attendance Reviewer	Monitor monthly sheets for missing IN/OUT records, geofence anomalies, and device anomalies; request corrections where needed.
Payroll Officer	Use monthly summary rows and grand totals for salary calculation; verify totals against employment contracts and policies.
System Administrator	Maintain Web App deployment, configure geofence center/radius, manage spreadsheet permissions, and ensure triggers are active.
Senior Software Developer	Handle escalations requiring backend changes (rate policy, anomaly rules, sheet structure updates).

5 SYSTEM REQUIREMENTS

Before recording attendance or reviewing wage data, confirm the following system requirements:

Table 3: System Requirements

Requirement	Details
Browser	Google Chrome (recommended), Microsoft Edge, Mozilla Firefox, or Safari - latest version.
Internet Connection	Stable connection required during IN/OUT punches and HR review.
Google Account	Organizational Google account with access to the Web App and edit access to the Target Sheet (for HR/Payroll).

Requirement	Details
Web App URL	Production attendance Web App URL provided by the System Administrator.
Target Google Sheet	Must exist, be linked in backend configuration, and be shared with HR/Payroll; monthly sheets are auto-created by the system.
Time Zone	System operates in IST (Asia/Kolkata); all times are stored in IST.
Location Services	Device must allow geolocation access for accurate GPS capture.

6 ATTENDANCE DATA REQUIREMENTS - WHAT IS RECORDED

The system writes structured data to the monthly sheets. Each attendance record contains the following fields:

Table 4: Attendance Data Requirements

#	Column	Purpose
1	Name	Employee name; must match Employees Sheet configuration.
2	Date	Attendance date in dd-MM-yyyy format (plain text).
3	Time	Time of IN/OUT in HH:mm:ss or user-edited format.
4	Status	IN or OUT.
5	Latitude	GPS latitude from browser.
6	Longitude	GPS longitude from browser.
7	Inside Geofence	YES/NO flag indicating whether location is within configured geofence.
8	DeviceId	Persistent device identifier used for anomaly detection.
9	Browser	Browser name and version.
10	OS	Operating system name and version.
11	Device Model	Device model derived from UA/Client Hints/fallback parsing.
12	Maps Link	Hyperlink to Google Maps location for audit purposes.
13	No. of Hours	Total hours between IN and OUT (human-readable, e.g., “8h 30m”).
14	Normal Hours	Hours up to the normal hours limit.
15	Overtime Hours	Hours exceeding the normal hours limit.
16	Total Wage	Calculated wage for this OUT record.

For wage calculation, each OUT must have a corresponding IN on the same date for the same employee, and the employee hourly rate must be configured in the Employees Sheet.

7 STEP-BY-STEP ATTENDANCE PUNCH INSTRUCTIONS

7.1 Accessing the Web Application

1. Open a supported browser (Google Chrome recommended).
2. Navigate to the attendance Web App URL provided by the System Administrator.
3. Sign in using your organizational Google account.
4. Confirm that the page displays:
 - Live clock showing date and time in IST.
 - Name dropdown.
 - IN and OUT buttons.

7.2 Selecting Your Name

1. Click on the Name dropdown.
2. Select your name from the list of active employees.
3. If your name does not appear, contact HR or the System Administrator. Do not proceed using another employee's name.

7.3 Granting Location Access

1. When prompted, allow location access in your browser.
2. If you deny location access, the system will show an error and attendance will not be recorded.

7.4 Punching IN

1. Ensure you are physically inside or near the office premises.
2. Select your name from the dropdown.
3. Click the **IN** button.
4. Wait while geolocation is fetched; the IN/OUT buttons will be temporarily disabled.
5. On success, confirm that a message such as “Submission recorded: IN” is displayed.
6. If an error occurs (location denied, system error), resolve the issue and retry.

7.5 Punching OUT

1. At the end of your working period, ensure you are within the geofence area where possible.
2. Select your name.
3. Click the **OUT** button.
4. Wait for the confirmation message “Submission recorded: OUT”.
5. The system will automatically calculate hours and wage for this OUT based on the corresponding IN.

8 STEP-BY-STEP WAGE CALCULATION AND REVIEW

8.1 Daily Wage Calculation Logic (Information)

For each OUT entry, the system:

1. Finds the most recent IN for the same employee on the same date.
2. Computes the time difference between OUT and IN in hours.
3. Rounds the duration to the nearest 5 minutes.
4. Splits hours into:
 - Normal Hours: up to the configured daily limit (e.g., 8 hours).
 - Overtime Hours: hours beyond this limit.
5. Calculates Total Wage:

$$\text{Normal Wage} = \text{Normal Hours} \times \text{Hourly Rate}$$

$$\text{Overtime Wage} = \text{Overtime Hours} \times \text{Overtime Rate}$$

(Note: Overtime rate is configured as 2× hourly rate).

6. Writes human-readable time strings and numeric wage into columns 13–16.

8.2 Reviewing Daily Entries (HR/Payroll)

1. Open the Target Google Sheet.
2. Navigate to the relevant monthly sheet.
3. Filter by employee name as required.
4. Confirm that:
 - IN and OUT pairs are correct and chronological.
 - Geofence status is reasonable (e.g., mostly YES).
 - Device anomalies are highlighted appropriately.
 - Wage values appear only in OUT rows.

8.3 Correcting IN/OUT Times

1. If an IN or OUT time needs adjustment, edit the time cell directly in the monthly sheet.
2. For OUT time edits, the system automatically recalculates wage for that row via the onEdit trigger.
3. For IN time edits, the system finds the corresponding OUT row and recalculates its wage.
4. After edits, verify that hours and wage columns reflect updated values.

9 STEP-BY-STEP MONTHLY SUMMARY AND REPORTING

9.1 Automatic Monthly Sheet Creation

1. When the first attendance record for a new month is submitted, the system:
 - Creates a new sheet named **MMM-yyyy** if it does not exist.
 - Adds header row and filter, and formats the sheet.
 - Snapshots active employees and their hourly rates for the month.

9.2 Monthly Summary Generation

1. After the month is complete, the system generates per-employee summary rows:
 - Each employee gets a row labeled **MONTHLY TOTAL** in the Date column.
 - A grand total row labeled **GRAND TOTAL** shows the overall wage sum.

9.3 Monthly Report Email

1. Once admin email is configured, the system:
 - Exports the monthly sheet as an XLSX file.
 - Sends the attendance report to the admin via email.
2. Payroll Officer can archive this file for salary processing.

10 EXCEPTION HANDLING

10.1 Attendance Punch Exceptions

E-01: Name not found in dropdown

- **Cause:** Employee not configured or inactive in Employees Sheet.
- **Action:** Contact HR/System Administrator to add or activate the record. Do not use another employee's name.

E-02: Geolocation not supported / denied

- **Cause:** Browser/device does not support geolocation or user denied permission.

- **Action:** Use a supported device/browser and allow location access; retry.

E-03: Web App not loading or showing error

- **Cause:** Network issues, Apps Script outage, or misconfigured deployment.
- **Action:** Check internet connection; escalate to System Administrator if persistent.

10.2 Data Validation Exceptions (Backend)

D-01: Error: No rate found for ;name;

- **Cause:** Hourly rate not configured or zero in Employees Sheet.
- **Action:** HR must set the hourly rate; wages will calculate thereafter.

D-02: Error: No IN Found for ;date;

- **Cause:** OUT record exists without matching IN.
- **Action:** HR should verify attendance; if IN was missed, add a manual IN or handle as per policy.

D-03: Error: IN time is after OUT time

- **Cause:** IN/OUT times are misordered.
- **Action:** Correct times so IN precedes OUT; wage recalculation will occur automatically.

10.3 Device and Geofence Exceptions

S-01: Device anomaly highlighted

- **Cause:** Device ID differs from established device within threshold.
- **Action:** HR review; if legitimate device change, system will accept new device after repeated usage; otherwise, investigate misuse.

S-02: Outside geofence

- **Cause:** GPS coordinates are outside the configured office radius.
- **Action:** HR review; apply policy (require explanation, adjust wage, etc.).

10.4 System and Access Exceptions

Invalid spreadsheet ID, permission denied, authorization prompts, or quota issues must be escalated to the System Administrator or Senior Software Developer for resolution.

11 DO'S AND DON'TS

11.1 DO's

Table 5: DO's

#	Do
1	Always punch IN at the start of your working period and OUT at the end.
2	Always ensure your name is correctly selected before punching.
3	Always allow location access and keep GPS enabled when using the Web App.
4	Always confirm that submission messages show "Submission recorded: IN/OUT".
5	HR and Payroll should always review monthly sheets and summary rows before salary processing.
6	System Administrator should always ensure Employees Sheet is up to date and triggers are active.
7	Always verify that wage calculations look reasonable after editing times.
8	Always report repeated device anomalies or outside-geofence records to HR.
9	Always use supported browsers (preferably Google Chrome).
10	Always escalate system errors promptly instead of manually altering backend formulas.

11.2 DON'Ts

Table 6: DON'Ts

#	Do Not
1	Do not punch attendance from unauthorized devices or locations purely to mark presence.
2	Do not use another employee's name for attendance.
3	Do not manually alter geofence or device anomaly highlighting in the sheet.
4	Do not edit wage formulas or overwrite calculated values in columns 13–16.
5	Do not disable location services when using the Web App.
6	Do not close the browser tab while an IN/OUT submission is in progress.
7	Do not share the Web App URL with unauthorized personnel.
8	Do not change monthly sheet names or Employees Sheet structure without consulting the System Administrator.
9	Do not ignore repeated anomalies or errors; always investigate or raise them.
10	Do not attempt to bypass the system by marking attendance directly in the sheet without using the Web App.

12 APPENDIX A - SHEET STRUCTURE AND COLUMN REFERENCE

Table 7: Sheet Structure and Column Reference

Column	Purpose
Name	Employee identifier.
Date	Attendance date in dd-MM-yyyy format (plain text).
Time	Punch time.
Status	IN/OUT.
Latitude	GPS latitude.
Longitude	GPS longitude.
Inside Geofence	YES/NO flag.
DeviceId	Device anomaly tracking.
Browser	Browser information.
OS	Operating system information.
Device Model	Device identification.
Maps Link	Quick location view in Google Maps.
No. of Hours	Total hours between IN and OUT.
Normal Hours	Hours up to daily limit.
Overtime Hours	Hours above daily limit.
Total Wage	Calculated wage per OUT.

13 APPENDIX B - EMPLOYEE CONFIGURATION AND RATE MANAGEMENT

Table 8: Employee Configuration

Column	Purpose
Name	Employee display name used in frontend dropdown.
Hourly Rate	Base hourly wage used for calculations.
Status	Active/inactive; only active employees appear in the Web App.

Monthly rate snapshots are taken when a new monthly sheet is created; changes made later affect future months but not past snapshots.

14 APPENDIX C - GEOFENCE AND DEVICE SECURITY RULES

Geofence Rules

- Office center is defined by latitude and longitude constants.
- Radius in meters defines allowable punch zone.
- Distance is calculated using the haversine formula; Inside Geofence is YES if distance is less than or equal to the radius.

Device Anomaly Rules

- Each employee has an established DeviceId stored in PropertiesService.
- A new DeviceId is considered anomalous until used for a configured number of consecutive punches.
- Anomalous entries are highlighted for HR review; once threshold is met, the new device is accepted as established.

15 REVISION HISTORY

Table 9: Revision History

Version	Date	Revised By	Change Description
1.0	June 22, 2026	Amanjyot Singh Bagga	Initial version - attendance, geofence, device anomaly, wage calculation, monthly summary, exception handling, and appendices.

This SOP must be reviewed whenever geofence parameters, employee rate policy, Target Sheet structure, or a new version of the attendance automation system is deployed.